Darshana Opticals (Pvt) Ltd

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Software Requirement Specification (SRS)

Version History

The following table provides a comprehensive record of the revisions made to this Software Requirement Specification, ensuring that all changes are documented for auditability and project tracking as required by the university standards for design projects.

Version No.	Date	Author(s)	Summary of Changes
0.1	2025/11/27	Project Team	Initial requirements gathering and concept definition for the initial University Presentation.
0.2	2026/01/15	Project Team	Refinement of system scope and high-level features for the Project Poster submission.

1.0	2026/03/21	Project Team	Final SRS document; integration of 21-section structure, feasibility studies, and detailed diagrams for final design approval.
1.1 (Review Candidate)	2026/10/24	Hiruni Imasha	Updated functional and non-functional requirement traceability, user roles, business rules, and system scope to reflect implemented functionality and accepted ADR-001–015. Incorporated deployment and testing evidence, documented known limitations, and identified pending verification activities.

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1. Introduction

This document presents the Software Requirements Specification (SRS) for the Optical Shop Management System (OSMS) developed for Darshana Opticals (Pvt) Ltd. The system supports the transition from manual optical retail and clinical operations to an integrated web-based platform.

This final SRS reflects the implemented system, approved architectural decisions, and known limitations while maintaining traceability to the original approved requirements.

2. Purpose

The primary purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements of the Optical Shop Management System (OSMS) developed for Darshana Opticals (Pvt) Ltd.

The document outlines the system's core functionalities, including online shopping, inventory management, appointment scheduling, clinical prescription management, loyalty point management, AI chatbot assistance, and administrative operations.

This final SRS reflects the implemented functionalities, approved architectural decisions (ADR-001 to ADR-015), and known system limitations while maintaining traceability to the original approved requirements.

Intended Audience: This document is intended for the following stakeholders:

- **Project Managers and Developers:** To provide a clear reference for system requirements, business rules, functional behavior, and implementation constraints.
- **Darshana Opticals Management:** To establish a shared understanding of the system's capabilities, approved changes, and remaining limitations.
- **Quality Assurance Teams:** To define the requirements and acceptance criteria used to verify system functionality, performance, security, and reliability.

3. Scope

System Name: The system is identified as the **Optical Shop Management System (OSMS)**.

What the System Does: The OSMS is a web-based platform developed to digitalize and manage the core business operations of Darshana Opticals (Pvt) Ltd. The following capabilities are provided:

- **Digital Presence:** Eyewear and accessories are showcased through an online product catalogue, allowing customers to browse, search, and purchase products using a shopping cart, secure payment integration, and order tracking.

- **Appointment Management:** Customers can book eye examinations through capacity-based Doctor Sessions at the approved Gampaha branch. Appointmentrelated communication is handled manually by authorized branch staff, as defined in ADR-007 and ADR-009.

- **Inventory Tracking:** Branch-specific inventory records are maintained in a centralized database, supporting stock updates, inventory monitoring, and automated low-stock alerts.

- **Customer Engagement:** Customer assistance is provided through an AI-driven informational chatbot. A loyalty points system automatically calculates and records eligible purchase-based rewards.

- **Clinical Prescription Management:** Authorized Optometrists can securely record customer prescription details, while customers can view their own prescription history.

- **Administration and Reporting:** The system provides role-based staff and Doctor management, sales and inventory reporting, and audit logging for relevant business operations.

Explicit Exclusions: The following areas are excluded from the current approved system scope:

- **Medical Diagnostic Hardware:** Clinical prescription information can be recorded and managed, but direct integration with medical diagnostic equipment is not supported.

- **Physical Delivery:** The system supports order status tracking, but physical delivery and fleet management are outside its scope. No automated courier-service integration is implemented.

- **Financial Auditing:** The system provides operational sales reporting and audit records, but full corporate financial auditing, accounting, and tax filing are not included.

- **Automated Appointment Reminders:** Automated email or SMS appointment reminders are excluded from the current approved scope under ADR-009. Appointment communication is handled manually.

Current Implementation Limitations: Loyalty point redemption, tier-based loyalty rewards, automated inter-branch stock transfers, and historical stock movement reporting are not available in the current implementation. Any original requirements not formally superseded remain traceable as outstanding requirements or limitations.

4. Definitions, Acronyms and Abbreviations

The following table defines the key terms, acronyms, and abbreviations used throughout this Software Requirements Specification (SRS) to ensure a common understanding among stakeholders.

Term	Definition
OSMS	**Optical Shop Management System** – The web-based system developed for Darshana Opticals (Pvt) Ltd.
SRS	**Software Requirements Specification** – A document describing the system's functional and non-functional requirements.
SDS	**Software Design Specification** – A document describing the system's architecture and technical design.
MERN	**MongoDB, Express.js, React, and Node.js** – The technology stack used to develop the system.
JWT	**JSON Web Token** – A token used to authenticate users and access protected system features.
SKU	**Stock Keeping Unit** – An identifier used in inventory management. The current OSMS uses database inventory identifiers rather than a shared global SKU.
AI	**Artificial Intelligence** – Technology used by the system's chatbot to provide informational assistance.
UI/UX	**User Interface / User Experience** – The system's visual interface and how users interact with it.
HTTP/HTTPS	**Hypertext Transfer Protocol / Hypertext Transfer Protocol Secure** – Protocols used for web communication; HTTPS provides encrypted communication.
API	**Application Programming Interface** – A set of rules that allows software components to communicate.
ERD	**Entity Relationship Diagram** – A diagram showing data entities and their relationships.
FR	**Functional Requirement** – A function or service the system must provide.
NFR	**Non-Functional Requirement** – A quality or constraint the system must satisfy, such as performance or security.
ADR	**Architectural Decision Record** – A document recording an approved technical or architectural decision and its justification.

RBAC	**Role-Based Access Control** – A security mechanism that restricts system access according to user roles.
Doctor Session	A scheduled examination period assigned to a Doctor, with a defined time and maximum patient capacity.
PDPA	**Personal Data Protection Act** – Sri Lankan legislation governing personal data protection.
PII	**Personally Identifiable Information** – Information that can identify an individual, such as a name or contact details.
TLS	**Transport Layer Security** – A protocol used to secure data transmitted over networks.
TTL	**Time To Live** – A mechanism used to automatically expire stored data, such as chatbot usage records.
IPN	**Instant Payment Notification** – A notification sent by PayHere to the backend to report a payment event for verification and processing.
CI/CD	**Continuous Integration / Continuous Delivery or Deployment** – Practices used to automate software validation and release processes.
UAT	**User Acceptance Testing** – Testing performed to determine whether the system meets approved user and business requirements.

Table 01: Definitions, Acronyms and Abbreviations

5. Document Overview

The remainder of this Software Requirements Specification (SRS) document is organized as follows:

- **Section 6 (Project Scope):** Describes the project's boundaries, business objectives, system capabilities, and exclusions.
- **Section 7 (Stakeholders):** Identifies the key stakeholders, including system administrators, branch staff, customers, and other relevant groups.
- **Section 8 (Functional Requirements):** Defines the system's functional requirements with unique identifiers, priorities, implementation status, and requirement traceability.
- **Section 9 (Non-Functional Requirements):** Specifies the system's quality attributes based on ISO/IEC 25010, including measurable performance, security, reliability, usability, and maintainability requirements.
- **Sections 10–12:** Describe the business rules, design constraints, and assumptions, considering approved changes and implementation limitations.
- **Sections 13–15:** Present use case diagrams, use case descriptions, and activity diagrams illustrating the system's main workflows.

- **Section 16 (External Interface Requirements):** Describes the user, hardware, software, and communication interface requirements.

- **Section 17 (Data Requirements):** Outlines the system's data requirements, storage, management, and security considerations.

- **Sections 18–20:** Present alternative solutions, technical, economic, operational, and schedule feasibility studies, and the project risk register.

- **Section 21 (References):** Lists the relevant documents, approved Architectural Decision Records (ADRs), and external references used in the SRS.

6. Project Scope

The Optical Shop Management System (OSMS) is developed to digitalize the core retail, clinical, and administrative operations of Darshana Opticals (Pvt) Ltd. The system provides a centralized web-based platform for managing customer services, branch inventory, appointments, orders, and business reporting.

The project scope is categorized into the following primary domains.

6.1 Retail and E-commerce Integration

A digital storefront is provided to display eyewear products, including frames, lenses, sunglasses, and related accessories. Customers can browse, search, and filter available products by category, brand, and price.

The system supports shopping cart management, online checkout, and secure payment processing through PayHere. Customers can also view their purchase history and track order status. Supported order status changes can trigger customer notifications.

6.2 Clinical Appointment Scheduling

An online appointment booking system allows customers to schedule eye examinations at the approved Gampaha Main Branch.

Appointments are managed through Doctor Sessions created by authorized staff. Each session has a defined schedule and patient capacity. Customers can view available sessions, select a Doctor, and book appointments at least 24 hours in advance.

Authorized branch staff can manage appointments and Doctor Sessions according to their assigned permissions.

Appointment-related communication is handled manually by authorized branch staff. Automated email and SMS appointment reminders are excluded from the current approved scope, as defined in ADR-007 and ADR-009.

6.3 Multi-branch Inventory Management

A centralized inventory database provides visibility into stock records maintained separately for each branch.

Authorized staff can manage product information, stock quantities, prices, and inventory images according to their roles and branch permissions.

The system automatically identifies low-stock conditions and maintains reorder alerts based on an approved organization-wide threshold.

Automated stock transfers between branches and historical inventory movement tracking are not supported in the current implementation.

6.4 Customer Relationship and Retention

The system maintains customer profiles containing relevant personal information and provides access to purchase history and clinical prescription history.

A loyalty points system automatically awards eligible points for qualifying purchases and maintains customer loyalty transaction records. Points are managed through the backend to ensure consistency and traceability.

Loyalty point redemption, tier-based rewards, and exclusive tier-based discounts are not implemented in the current release.

An AI-driven chatbot provides informational assistance to authenticated customers through controlled usage limits and approved knowledge sources.

6.5 Administrative Operations and Reporting

The system provides administrative functions for authorized staff and Doctor management, along with role-based access to business operations.

Sales reporting tools allow authorized management users to review payment-based sales performance, order statistics, and relevant business summaries.

Inventory reporting provides current stock information, branch-level stock visibility, and low-stock or reorder-related statistics.

System Administrators can access approved organization-wide reports, while Branch Managers are restricted to authorized branch-level reporting.

The system also maintains audit records for relevant business and administrative activities.

Historical inventory turnover analysis and dedicated appointment-frequency analytics are not included in the current reporting implementation.

6.6 Clinical Prescription Management

The system supports the recording and management of clinical optical prescriptions by authorized Optometrists.

Prescription records are associated with the relevant customers and contain the supported examination details.

Customers can securely view their own prescription history, while authorized clinical staff can access prescription information according to their assigned permissions.

Direct integration with medical diagnostic equipment is outside the project scope.

7. Stakeholders

The key individuals and groups involved in or affected by the Optical Shop Management System (OSMS) are identified in this section. Their roles and primary interests are summarized in the following table, reflecting the implemented system and the approved role model defined in ADR-002.

Stakeholder	Category	Role and Interest
System Administrator	Internal	Manages authorized administrative operations, including staff accounts, Doctor records, user-role assignments, and system-wide business reporting. Access is restricted according to the approved role-based permissions.
Branch Manager	Internal	Oversees assigned branch operations, manages inventory and reorder alerts, monitors branch-specific sales and inventory reports, and supervises operational activities.
Branch Assistant	Internal	Handles daily branch operations, customer inquiries, order processing and tracking, Doctor Session coordination, and appointment management according to authorized permissions.

Optometrist	Internal	Records clinical optical prescriptions, accesses authorized customer prescription history, and supports clinical examination activities.
Doctor	Clinical Service Provider	Conducts scheduled eye examinations through assigned Doctor Sessions. Doctor information is maintained in a separate directory, and Doctors do not have a dedicated OSMS login role.
Customers	External	Browse and purchase eyewear products, manage their profiles, book appointments, view prescription and purchase history, track orders, access loyalty information, and use the AI chatbot.
Darshana Opticals Management / Owners	Internal / Owner	Oversee business operations and use authorized sales and inventory reports to support operational and strategic decisions. System access depends on an assigned approved role.
Payment Service Provider (PayHere)	External	Provides the payment gateway used for online transactions and sends payment notifications to the backend for verification and processing.
External Technology Service Providers	External	Provide supporting services, including Cloudinary for inventory image storage and an approved AI service for chatbot functionality.
Third-party Logistics Providers	External / Outside Current Integration Scope	May handle physical product delivery independently of OSMS. Automated courier integration and delivery-status synchronization are not implemented in the current system.

Table 02: Stakeholders of the Optical Shop Management System

8. Functional Requirements

The fundamental services and specific behaviors of the Optical Shop Management System (OSMS) are defined in this section. Each requirement is uniquely identified and prioritized according to the original approved SRS.

The implementation status of each requirement is recorded based on the current system functionality, accepted Architectural Decision Records (ADRs), and identified limitations.

ID	Description	Priority	Source / Stakeholder	Status
FR-001	The system provides secure customer registration and authentication. Staff accounts are provisioned through authorized administrative operations, with access controlled by assigned user roles.	High	System Administrator / Customer	**Implemented**
FR-002	Customers can manage their personal profile information and securely view their own clinical prescription history.	Medium	Customer	**Implemented**
FR-003	The system maintains a digital product catalogue, allowing customers to browse, search, and filter eyewear products by category, brand, and price.	High	Branch Manager	**Partially Implemented**
FR-004	The system provides a persistent shopping cart and secure online checkout process, including order creation and inventory reservation.	High	Customer	**Implemented**
FR-005	The system maintains centralized, branch-specific inventory records and updates stock quantities through authorized inventory and order operations.	High	Branch Manager	**Partially Implemented**
FR-006	The system automatically generates and maintains low-stock alerts when inventory quantities reach the configured organization-wide reorder threshold.	Medium	Branch Manager	**Implemented**

FR-007	Customers can book eye examination appointments through available, capacity-based Doctor Sessions at the approved Gampaha Main Branch, subject to the minimum booking lead time and approved scheduling rules.	High	Customer / Branch Assistant	**Modified by Approved Decision**
FR-008	Appointment-related customer communication is handled manually by authorized branch staff. Automated appointment email and SMS reminders are excluded from the current approved scope under ADR-009.	Medium	Branch Assistant	**Modified by Approved Decision**
FR-009	The system integrates PayHere to support secure online card payment processing, including payment initiation and backend verification of payment notifications.	High	Branch Manager / System Administrator	**Implemented**
FR-010	The system automatically calculates and records loyalty points for eligible customer purchases, maintaining an authoritative customer balance and transaction history.	Low	Customer / Branch Manager	**Implemented**
FR-011	The system provides an AI-driven chatbot to assist authenticated customers with general information and inquiries, subject to approved usage limits and knowledge-source restrictions.	Low	Customer	**Implemented**
FR-012	The system generates role-restricted sales reports and current inventory reports for authorized management users, including relevant summaries, filters, and exports.	High	System Administrator / Branch Manager	**Implemented**
FR-013	Authorized Optometrists can securely record and store clinical prescription information, with controlled access to customer prescription history.	High	Optometrist	**Implemented**
FR-014	The system allows customers to track their orders through the approved order lifecycle, with status updates performed by authorized branch staff and relevant customer notifications.	Medium	Customer / Branch Assistant	**Modified by Approved Decision**

Table 03: Functional Requirements of the Optical Shop Management System Implementation Status and Traceability Notes

FR-003 — Product Catalogue: Product searching and filtering by category, brand, and price are implemented. The original requirement also mentioned style-based filtering; a separate style filter has not been verified in the current implementation. This remains a requirement gap unless formally superseded.

FR-005 — Inventory Tracking: Centralized branch-specific stock records and inventory updates are implemented. However, the original real-time synchronization wording is broader than the behavior independently verified. A dedicated live synchronization mechanism and inter-branch stock-transfer workflow have not been established as implemented.

FR-007 — Appointment Booking: The original appointment model was modified through ADR-007 to use capacity-based Doctor Sessions. The current appointment scope is limited to the Gampaha Main Branch, with bookings made at least 24 hours before the session start.

FR-008 — Appointment Reminders: The original automated reminder requirement is not implemented. ADR-009 formally approves manual appointment communication for the current scope. This is an approved requirement deviation, not evidence that automated reminders have been delivered.

FR-010 — Loyalty Points: Automatic earning for eligible purchases is implemented. Loyalty redemption and tier-based rewards are not implemented. Their absence must remain documented separately rather than being concealed by the implemented accrual status.

FR-014 — Order Tracking: ADR-004 defines the implemented order lifecycle as ORDER_CREATED, PROCESSING, READY_FOR_PICKUP, PICKED_UP, COMPLETED, and CANCELLED. The system does not provide third-party physical delivery tracking or a live courier integration.

Verification Boundary: Functional implementation statuses do not independently establish final system acceptance or compliance with every non-functional requirement. DDP-084 system integration and E2E testing and DDP-085 security verification have been completed, with their respective evidence documented. DDP-087 final test case register and consolidated QA evidence remain subject to separate completion and approval. Any unverified acceptance criteria continue to be identified in the applicable requirement-status records.

9. Non-Functional Requirements

The quality attributes and operational constraints of the Optical Shop Management System (OSMS) are specified in this section. These requirements define the expected standards for system performance, reliability, usability, security, scalability, maintainability, and portability.

The original approved acceptance criteria are preserved. Current implementation and testing evidence are summarized separately to distinguish verified results from outstanding requirements.

Category	Requirement ID	Description	Measurable Acceptance Criterion
Performance	**NFR-001**	System responsiveness is maintained under standard operational loads.	95% of database search queries shall be processed and displayed within 2 seconds.
Reliability	**NFR-002**	Continuous availability of the system is ensured for business operations.	An uptime of 99.9% shall be maintained, excluding scheduled maintenance periods.
Usability	**NFR-003**	The user interface is designed for ease of use by nontechnical staff.	New staff members shall be able to process a complete sales transaction within 10 minutes of initial training.
Security	**NFR-004**	Unauthorized access to sensitive customer and prescription data is prevented.	All data transmissions shall be encrypted using SSL/TLS, and passwords shall be hashed using Bcrypt.
Scalability	**NFR-005**	The system architecture is designed to handle business growth.	The system shall support a 50% increase in concurrent users and transactions without a degradation in response time.

Maintainability	**NFR-006**	The codebase is structured to allow for efficient updates and bug fixes.	A modular architecture is utilized, ensuring that 90% of minor bug fixes are implemented and deployed within 24 hours.
Portability	**NFR-007**	Cross-platform compatibility is provided for various user environments.	The web application shall be fully functional across the latest versions of Google Chrome, Mozilla Firefox, and Safari.

Table 04: Non-Functional Requirements of the Optical Shop Management System

9.1 Non-Functional Requirement Verification Status

The following table summarizes the available implementation and testing evidence. A requirement is considered fully verified only when the evidence demonstrates its original measurable acceptance criterion.

Requirement ID	Verification Status	Evidence and Remaining Verification
NFR-001	Partially Verified	Local backend performance tests met a p95 API response-time threshold of 2 seconds across 13 critical endpoints. Full databasesearch-to-browser-display timing under normal hosted conditions remains unverified.
NFR-002	Not Verified	Local load and reliability tests reported no server crashes or unexpected request failures. However, sustained hosted availability of 99.9% has not been demonstrated.
NFR-003	Not Verified	The required staff usability assessment and 10-minute transaction-completion criterion have not been demonstrated through recorded user testing.
NFR-004	Partially Verified	Bcrypt password hashing, JWT authentication, and role-based access controls are implemented. DDP-085 security verification has been completed, including

		OWASP assessment, targeted security hardening, and deployed HTTPS/HSTS verification for the Vercel frontend and Render backend. No unresolved Critical or High dependency vulnerabilities were reported at the verified assessment baseline. Residual Moderate dependency risks remain documented. Complete verification of every transport and third-party integration boundary is not established; therefore, the requirement remains Partially Verified.
NFR-005	Not Fully Verified	Controlled local load tests evaluated 5, 15, and 30 virtual users. The tests did not establish the original requirement of a 50% user and transaction increase without response-time degradation.
NFR-006	Not Fully Verified	The system uses a modular architecture with automated tests and CI workflows. Evidence showing that 90% of minor fixes were implemented and deployed within 24 hours is not available.
NFR-007	Not Verified	Browser compatibility testing across the latest Chrome, Firefox, and Safari versions has not been established through documented evidence.

Table 05: Non-Functional Requirement Verification Status

9.2 Performance and Reliability Evaluation Summary

Performance and reliability testing was conducted under DDP-086 using isolated local test environments and synthetic data.

The evaluation reported that all 13 critical backend endpoint benchmarks achieved p95 response times below 130 ms. Under a controlled workload of 30 concurrent virtual users, the recorded p95 API response time was 241.78 ms, with no unexpected request failures in the evaluated scenarios.

The performance assessment also verified transaction safety under concurrent inventory checkout and Doctor Session booking operations.

These findings provide evidence of local backend performance and reliability. They do not independently establish production availability, complete browser response times, or compliance with every original non-functional acceptance criterion.

Final hosted integration, security, usability, and cross-browser verification must be recorded through the applicable testing and QA activities.

Completed Verification References: DDP-084 (System Integration and E2E Testing), DDP-085 (Security Verification), and DDP-086 (Performance and Reliability Testing).

Outstanding Documentation Reference: DDP-087 (Final Test Case Register and Consolidated QA Evidence).

10. Business Rules

The operational policies and decision-making constraints of Darshana Opticals (Pvt) Ltd are defined in this section. These rules guide system behavior and support consistency across business operations.

The current implementation follows the approved Architectural Decision Records (ADRs). Original business rules that have not been implemented or independently verified are retained and identified as outstanding requirements.

10.1 Appointment Management Rules

- **Booking Lead Time:** Customers must book eye examination appointments at least 24 hours before the selected Doctor Session begins.

- **Cancellation Policy:** Customers may cancel their confirmed appointments before the scheduled Doctor Session starts. The original 12-hour cancellation restriction has been superseded by ADR-007.

- **Doctor Session Allocation:** Appointments are managed through staff-created Doctor Sessions with defined starting times, ending times, and maximum patient capacities. Multiple patients may book the same session, provided its capacity is not exceeded.

- **Branch Restriction:** Online eye examination appointments are available only at the approved Gampaha Main Branch under the current scope.

- **Appointment Communication:** Appointment-related communication is handled manually by authorized branch staff. Automated appointment email and SMS reminders are excluded from the current approved scope under ADR-009.

10.2 Inventory and Sales Rules

- **Stock Re-ordering:** The system identifies inventory items as low stock when their quantity is equal to or below the configured organization-wide reorder threshold. The approved default threshold is five units. Low-stock alerts are

generated and maintained automatically; automatic supplier purchase-order creation is not provided.

- **Discount Application:** The original approved business rule states that seasonal or promotional discounts must not be applied to items already classified as Clearance stock. Automated enforcement of this rule has not been verified in the current implementation.
- **Prescription Validity:** The original approved business rule requires prescription lens orders to be associated with an eye examination performed within the preceding 24 months. Automated validation of this condition during checkout has not been verified in the current implementation.

10.3 Loyalty and Customer Rules

- **Point Accrual:** Customers earn loyalty points at a rate of one point for every LKR 100 of eligible merchandise spending. Clinical eye examination fees are excluded. Points are calculated using authoritative order and payment information and awarded when the qualifying paid order reaches the COMPLETED status.

- **Point Redemption:** The original approved business rule permits redemption only when a customer's loyalty balance exceeds 500 points. This rule remains documented, but loyalty point redemption is not implemented in the current release.

- **Return Policy:** The original approved business rule permits returns of eyewear frames within seven days of purchase, provided the items remain in their original condition and are accompanied by a valid receipt. Automated enforcement of this return policy is not implemented in the current system.

10.4 User Access Rules

- **Managerial Override:** Branch Managers are authorized to perform inventory adjustments and price updates within their approved branch scope. System Administrators may perform authorized price-only overrides but do not have unrestricted inventory adjustment permissions. Access is enforced through the approved role-based authorization model.

- **Session Security:** The original approved requirement specifies automatic termination of user sessions after 30 minutes of inactivity. The current implementation uses JWT-based authentication and token expiration, but a separate 30-minute inactivity timeout has not been verified. This requirement remains outstanding.

10.5 Outstanding Business Rule Implementation

- The original business rules concerning promotional discount restrictions, prescription age validation, loyalty point redemption, eyewear frame returns, and automatic session termination after 30 minutes of inactivity are not confirmed as fully enforced in the current implementation.

- These rules are retained for requirements traceability and must not be presented as implemented functionality.

- The current system also records verified manual full-refund evidence and associated loyalty reversals. This functionality does not constitute an automated customer return-management workflow or establish enforcement of the original seven-day return policy.

- The outstanding rules and applicable approved deviations must be considered during final requirement reconciliation, security verification, and acceptance testing.

11. Constraints

The design, implementation, and operation of the Optical Shop Management System (OSMS) are governed by technical, regulatory, and operational constraints. These constraints define the system's development boundaries, compliance expectations, and deployment limitations.

11.1 Technical Constraints

- **Technology Stack:** The system is developed using the MERN technology stack (MongoDB, Express.js, React, and Node.js), in accordance with the approved application architecture.

- **Web Browser Compatibility:** The web application is required to support the latest versions of Google Chrome, Mozilla Firefox, and Safari, as specified in NFR-007. Microsoft Edge may be supported as an additional browser, subject to compatibility verification.

- **Hosting Environment:** The system requires cloud-based hosting. The current deployment uses Vercel for the React frontend, Render.com for the Node.js/Express backend, and MongoDB Atlas for database hosting. Render.com was adopted as an approved alternative to the originally planned AWS backend hosting approach, as documented in ADR-015.

- **Connectivity:** An internet connection is required to access the hosted application and centralized database services. Offline sales processing and synchronization after offline operation are not supported in the current implementation.

11.2 Regulatory and Legal Constraints

- **Data Privacy:** Personal and clinical prescription data must be handled in accordance with applicable provisions of Sri Lanka's Personal Data Protection Act (PDPA). The system must restrict access to sensitive information through appropriate authentication, authorization, and data protection controls.

- **Financial Standards:** Online payment processing is performed through the integrated PayHere payment gateway. Payment integration must meet applicable PCI DSS requirements. The OSMS does not directly store customers' payment card credentials. Independent verification of the payment provider's current certification and overall compliance remains necessary.

- **Audit Trails:** Critical financial transactions, inventory adjustments, and other approved auditable actions must be recorded in protected audit logs. The approved minimum retention requirement is seven years. The current implementation provides application-level immutable, append-only audit records without automatic TTL deletion. Long-term operational retention, backup, and archival guarantees remain subject to deployment and compliance verification.

11.3 Operational and Design Constraints

- **Language Support:** The current user interface supports English. Sinhala and Tamil language interfaces are excluded from the present development scope.

- **User Interface Consistency:** The application interface must follow the approved Darshana Opticals (Pvt) Ltd branding and maintain consistent use of colors, typography, and interface components.

- **Budgetary and Schedule Limits:** The project was originally planned within an allocated development budget and a six-month delivery schedule. Subsequent implementation and deployment decisions are documented through the project's issue tracking and architectural decision records. Completion within the original timeline is not asserted without supporting project evidence.

- **Implementation and Verification Note:** Browser compatibility, formal data protection compliance, payment-provider certification, seven-year operational audit retention, and branding conformity require supporting verification evidence. These constraints must not be treated as fully satisfied solely because the relevant architecture or application features exist.

12. Assumptions

The development and operation of the Optical Shop Management System (OSMS) are based on the following assumptions. These conditions are considered necessary for the effective implementation and use of the system but are not treated as verified facts unless supported by appropriate evidence.

12.1 Technical and Infrastructure Assumptions

- **Hardware Availability:** It is assumed that the physical branches of Darshana Opticals (Pvt) Ltd have access to suitable computing devices, such as desktop computers or tablets, and reliable internet connectivity for daily system operations.

- **Server Performance:** The cloud hosting services used by the system, including Vercel, Render.com, and MongoDB Atlas, are assumed to provide sufficient resources for normal business operations. Actual performance and availability remain subject to infrastructure limitations and verification against the approved Non-Functional Requirements.

- **Third-party APIs:** External services, including PayHere for payment processing, Cloudinary for image storage, and the approved AI service for chatbot functionality, are assumed to remain available during normal operations. Relevant email services are also assumed to be available where required. Temporary service interruptions may affect dependent functionality. Automated SMS appointment reminders are not part of the current approved scope.

12.2 Data and User Assumptions

- **Data Accuracy:** Existing manual records, including inventory information and customer prescription history, are assumed to be accurate or appropriately validated before any migration into the digital database. Completion of historical data migration is not assumed without supporting evidence.

- **User Competency:** Staff members of Darshana Opticals are assumed to possess basic computer literacy and receive appropriate training to operate the system effectively.
- **Customer Access:** The target customer base is assumed to have reasonable access to smartphones or computers with internet connectivity to use the online store, appointment booking, and other customer-facing services.

12.3 Organizational and Legal Assumptions

- **Stakeholder Availability:** Timely feedback, requirement clarification, and necessary approvals from Darshana Opticals management and other relevant stakeholders are assumed to be available during system validation and User Acceptance Testing (UAT).

- **Regulatory Stability:** The original project planning assumed that no major changes to applicable data protection and financial regulations would require significant architectural modifications during the planned six-month development period. Any relevant regulatory changes must be reviewed to determine their impact on the implemented system and its compliance obligations.

13. Use Case Diagrams

The functional interactions between the identified actors and the Optical Shop Management System (OSMS) are represented in Figure 01.

The updated use case diagram reflects the implemented system functionalities, approved rolebased access control model, and relevant architectural decisions. It illustrates how customers, authorized staff members, administrators, and external payment services interact with the system.

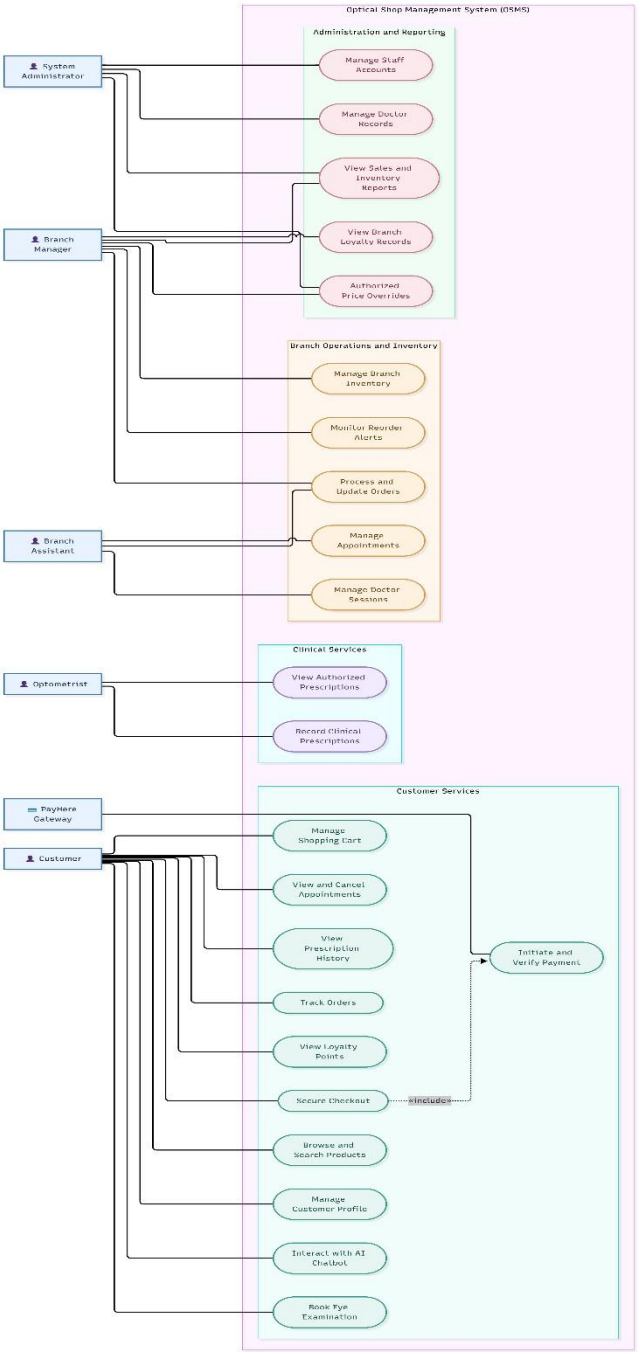


Figure 01: Use Case Diagram of the Optical Shop Management System (OSMS)

13.1 Identified Actors

The system supports five primary authenticated user roles and an external payment service:

- **Customer:** Interacts with the system to browse and purchase eyewear, manage personal information, book appointments, view prescription history, track orders, check loyalty points, and access the AI chatbot.

- **Branch Assistant:** Manages Doctor Sessions, coordinates appointments, handles customer inquiries, and performs authorized order-processing activities.

- **Branch Manager:** Oversees branch operations, manages inventory, monitors reorder alerts, updates authorized order statuses, and accesses branch-specific business reports.

- **Optometrist:** Records clinical prescription information and accesses authorized customer prescription history.

- **System Administrator:** Manages staff accounts and Doctor records, accesses authorized organization-wide reports, and performs permitted administrative operations.

- **PayHere Payment Gateway:** An external service that processes online payments and sends payment notifications for secure backend verification.

13.2 Use Case Relationships

The diagram uses standard UML relationships to represent interactions between actors and system functions:

- **Association:** Solid lines connect actors to the use cases they are authorized to perform.

- **«include»:** Mandatory supporting functionality is represented using include relationships where appropriate. For example, the secure checkout process includes order creation, inventory reservation, and payment initiation.

- **«extend»:** Optional behavior may be represented using extend relationships where appropriate. For example, filtering products can extend the product-browsing process.

- **Authentication:** Protected functions require authenticated users with the appropriate permissions. Public product browsing does not require authentication.

The diagram excludes unsupported functions such as automated appointment reminders, loyalty point redemption, automated human-agent escalation, and third-party delivery tracking.

13.3 Functional Grouping

The system's capabilities are organized into four logical functional areas:

- **Customer Services:** Includes product browsing, shopping cart management, checkout, appointment booking, order tracking, customer profiles, loyalty information, and AI chatbot assistance.

- **Branch Operations and Inventory:** Includes branch inventory management, reorder alerts, Doctor Session management, appointment coordination, and order processing.

- **Clinical Services:** Includes secure prescription recording and authorized prescriptionhistory access.

- **Administration and Reporting:** Includes staff management, Doctor record management, authorized price overrides, and role-restricted sales and inventory reporting.

These functional areas represent the approved scope of the implemented OSMS and maintain consistency with the functional requirements described in Section 8.

14. Use Case Descriptions

Detailed descriptions of the primary use cases identified in the previous section are provided here. Each description outlines the actors, preconditions, post-conditions, main success scenario, and alternative flows based on the implemented system and approved architectural decisions.

14.1 Use Case: Book Eye Test (UC-03)

Field	Description
Use Case ID	UC-03
Use Case Name	Book Eye Test
Actors	Customer
Description	The customer books an eye examination by selecting an available Doctor and a capacity-based Doctor Session at the Gampaha Main Branch.

Preconditions	1. The customer is logged into a valid OSMS account. 2. At least one Doctor Session with available capacity exists. 3. The selected session starts at least 24 hours after the booking request.
Postconditions	1. A new appointment is created with CONFIRMED status. 2. The appointment is associated with the selected Doctor Session, and the confirmed booking count is updated. 3. The customer receives an appointment reference and an assigned queue position. 4. The booking confirmation is displayed within the system.

Main Success Scenario (Flow of Events):

1. The customer selects the **Book Eye Test** option.
2. The system displays the customer's profile information and the approved Gampaha Main Branch.
3. The customer reviews their details and proceeds to Doctor selection.
4. The system displays the available Doctors.
5. The customer selects a preferred Doctor.
6. The system displays the available dates and Doctor Sessions for the selected Doctor.
7. The customer selects a suitable date and Doctor Session.
8. The system displays the selected appointment details for review.
9. The customer confirms and submits the booking request.
10. The system validates the booking lead time and available session capacity.
11. The system creates the appointment with CONFIRMED status and assigns a queue position.
12. The system displays the booking confirmation and appointment reference.

Alternative Flows:

- **A1: Session Fully Booked:** If the selected Doctor Session reaches its maximum capacity before confirmation, the system rejects the booking and prompts the customer to select another available session.

- **A2: Booking Lead Time Violation:** If the selected session starts within the required 24hour booking period, the system rejects the request and prompts the customer to select an eligible session.

- **A3: No Available Sessions:** If the selected Doctor has no available sessions, the system informs the customer and allows another Doctor or available date to be selected.

- **A4: Invalid Session:** If the selected session is no longer available or its details have changed, the system displays an error and refreshes the available sessions.

- **A5: Invalid or Incomplete Profile Information:** If the customer's profile details require correction, the customer can navigate to the profile page to update the relevant information.

Note: Appointment communication is handled manually by authorized branch staff.
Automated appointment reminders are not part of the current approved system scope.

14.2 Use Case: Manage Multi-branch Inventory (UC-05)

Field	**Description**
Use Case ID	UC-05
Use Case Name	Manage Multi-branch Inventory
Actors	Branch Manager (Primary), System Administrator (Limited price-override access)

Description	Authorized users view and maintain centralized, branch-specific inventory records for eyewear products. Branch Managers manage inventory information and stock quantities within their assigned branches, while System Administrators have restricted inventory-viewing and price-override permissions.
Preconditions	1. The user is authenticated with an authorized role. 2. The relevant branch and inventory records are available in the system. 3. The Branch Manager has permission to manage inventory within the assigned branch.
Postconditions	1. Authorized inventory changes are saved in the centralized database. 2. Updated stock quantities are reflected in the relevant branch inventory records. 3. Low-stock alerts are created, updated, or resolved according to the configured organization-wide reorder threshold. 4. Relevant inventory modifications are recorded in the audit trail.

Main Success Scenario (Flow of Events):

1. The Branch Manager logs into the OSMS and accesses the Inventory Management module.
2. The system verifies the user's role and assigned branch.
3. The system retrieves and displays the inventory records associated with the authorized branch.
4. The Branch Manager selects an existing inventory item for modification or chooses to create a new inventory item.
5. The Branch Manager enters or updates the relevant product information and stock quantity.
6. The system validates the submitted data and the user's authorization.
7. The system saves the authorized changes in the centralized database.
8. The system evaluates the updated stock quantity against the configured organizationwide reorder threshold.
9. If the stock quantity is at or below the threshold, the system creates or updates the relevant low-stock alert.
10. The system displays the updated branch inventory information and confirms the successful operation.

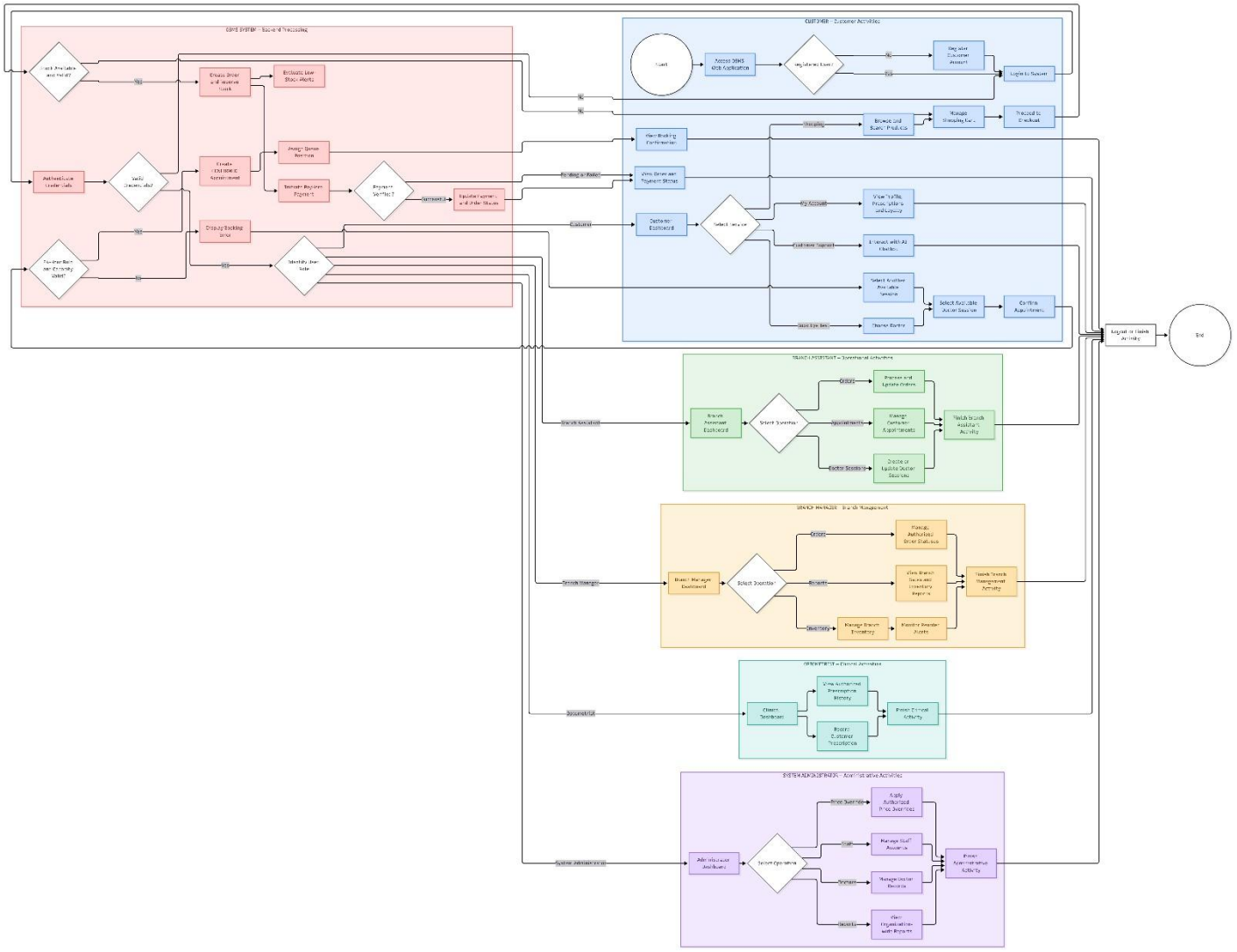
Alternative Flows:

- **A1: Unauthorized Branch Access:** If the user attempts to access or modify inventory outside their authorized branch, the system denies the operation.
- **A2: Invalid Inventory Data:** If required information is missing or invalid, the system rejects the request and displays an appropriate validation error.
- **A3: Inventory Item Not Found:** If the selected inventory item no longer exists, the system displays an error and allows the user to return to the inventory list.
- **A4: System Administrator Price Override:** A System Administrator may select an authorized inventory record and update its price. The system rejects attempts to modify stock quantities or other restricted inventory fields through this permission.
- **A5: Reorder Alert Resolution:** If the updated stock quantity exceeds the configured reorder threshold, the system resolves any applicable active low-stock alert.

15. Activity Diagrams

The activity flow of the Optical Shop Management System (OSMS) is illustrated in this section. The diagrams represent the sequence of activities, decision points, and interactions between the system and its authorized users.

The updated activity flows reflect the implemented functionalities and approved architectural decisions.



15.1 Overall System Process Flow

Figure 02: Overall System Activity Diagram of the Optical Shop Management System (OSMS)

The overall activity diagram presents the main operational processes of the OSMS. The activities are grouped according to the responsibilities of Customers, Branch Assistants, Branch Managers, Optometrists, System Administrators, and the OSMS backend.

- **Customer Path:** Customers can browse products, manage shopping carts, complete online purchases, book eye examinations, track orders, access prescription and loyalty information, and use the AI chatbot. Authentication is required for protected customer functions.

- **Branch Assistant Path:** Branch Assistants manage Doctor Sessions, coordinate appointments, and process customer orders according to their assigned permissions.

- **Branch Manager Path:** Branch Managers oversee branch inventory, monitor reorder alerts, process authorized order status changes, and access branch-specific sales and inventory reports.

- **Optometrist Path:** Authorized Optometrists record clinical prescription information and view permitted prescription histories.
- **System Administrator Path:** System Administrators manage staff accounts and Doctor records, access organization-wide reports, and perform authorized price overrides.

- **System Processing:** The backend validates authentication and permissions, processes inventory reservations, verifies payment notifications, validates Doctor Session availability, creates appointments, and evaluates reorder conditions.

15.2 Detailed Activity: Online Appointment Booking

The following steps describe the activity flow for booking an eye examination:

1. The customer logs into the OSMS and opens the **Book Eye Test** module.
2. The system displays the customer's profile information and the Gampaha Main Branch as the appointment location.
3. The customer selects an available Doctor from the provided list.
4. The system displays the available dates and capacity-based Doctor Sessions for the selected Doctor.
5. The customer selects a preferred date and Doctor Session and reviews the appointment details.
6. The customer submits the booking request.
7. **Decision Node:** The backend verifies that the selected session is valid, begins at least 24 hours after the booking request, and has remaining patient capacity.
8. **If Valid:** The system creates an appointment with CONFIRMED status, updates the session's confirmed booking count, and assigns a queue position.

9. **If Invalid:** The system displays an appropriate error message and allows the customer to select another available Doctor Session.
10. The system displays the successful booking confirmation, including the Doctor, session date and time, appointment reference, and queue position.

Appointment-related communication is handled manually by authorized branch staff. Automated appointment reminders are not included in the current approved implementation.

16. External Interface Requirements

The requirements for interfaces between the Optical Shop Management System (OSMS) and external entities, including users, hardware devices, and third-party software services, are specified in this section. These requirements reflect the implemented system architecture and approved interface boundaries.

16.1 User Interface Requirements

- **Responsiveness:** The system provides a web-based user interface developed using React.js, designed to support access from desktop computers, tablets, and smartphones.

- **Ease of Navigation:** The user interface is designed to provide clear and intuitive navigation for customers and staff, including users with limited technical experience. The system supports straightforward access to product browsing, appointment booking, and authorized management functions.

- **Accessibility Standards:** The user interface should follow applicable Web Content Accessibility Guidelines (WCAG) to support accessible navigation and interaction. Accessibility features are incorporated into relevant interface components; however, full WCAG compliance has not been independently verified. The chatbot uses an internal knowledge source to provide informational assistance rather than a separate customer-facing Knowledge Area portal.

- **Consistency:** A clean and professional interface is maintained throughout the application to reflect the branding and visual identity of Darshana Opticals (Pvt) Ltd.

16.2 Hardware Interface Requirements

- **Computing Devices:** The system can be accessed through standard computing devices equipped with a modern web browser and a stable internet connection. No specialized hardware is required for normal web-based operations.

- **Peripheral Support:** Direct integration with retail peripherals, such as barcode scanners and receipt printers, is not implemented in the current system. Inventory and order management are performed through the web interface.

- **Camera Integration:** Direct webcam integration and virtual eyewear try-on functionality are excluded from the current implementation.

16.3 Software Interface Requirements

- **Database Management:** The Node.js/Express.js backend interfaces with MongoDB Atlas for the storage and retrieval of customer information, inventory records, appointments, clinical prescriptions, orders, and other business data.

- **Payment Gateway:** The system integrates with PayHere to support online payment processing. Payment initiation is handled through the backend, and server-to-server payment notifications are verified before the corresponding payment status is updated.

- **Messaging Services:** The system provides SMTP-based email notification functionality using Nodemailer for supported operations, including password recovery and relevant order-status notifications. Email delivery depends on the configured messaging service. Automated SMS and email appointment reminders are excluded from the current scope under ADR-009.

- **Image Storage:** Cloudinary is integrated to support secure upload, storage, and management of inventory product images.

- **AI Chatbot Service:** The system integrates with the configured Gemini AI service to provide knowledge-based informational assistance to authenticated customers.

- **Operating Systems:** The web application is designed to operate through modern browsers on common operating systems, including Windows, macOS, and Linux. Cross-platform and browser compatibility remain subject to final verification.

16.4 Communications Interface Requirements

- **Communication Protocol:** Communication between the deployed React frontend and Node.js/Express backend is conducted over HTTPS to protect data during transmission.

- **Architectural Style:** The frontend communicates with the backend through RESTful APIs. Authentication and role-based authorization are applied to protected endpoints.

- **Data Format:** JSON (JavaScript Object Notation) is the primary format used for communication between the frontend and backend. Multipart form data is used for supported inventory image uploads, while PayHere payment notifications use formencoded data.

- **Network Standards:** Standard TCP/IP networking is used to support communication between users, the hosted application, backend services, and external providers. Database access is handled through the backend rather than through direct browser connections.

- **External Service Communication:** The backend communicates with external services, including PayHere, Cloudinary, Gemini AI, and configured email services, using their supported interfaces. Service availability and successful communication depend on external providers and deployment configuration.

17. Data Requirements

The organization, storage, and management of information within the Optical Shop Management System (OSMS) are defined in this section. The system uses a centralized database to support consistency, security, and controlled access to operational information across Darshana Opticals (Pvt) Ltd.

17.1 Database Management System

- **Non-relational Structure:** MongoDB is used as the document-oriented database management system. Mongoose schemas and validation rules are applied to maintain structured records, data consistency, and relationships between system entities.

- **Centralized Storage:** Operational data, including customer profiles, branch inventory records, Doctor Sessions, appointments, prescriptions, orders, and payments, is stored in MongoDB Atlas. Authorized users access this data through the backend APIs.

17.2 Data Entities and Relationships

The system uses related MongoDB collections to represent its core business entities. The Entity Relationship Diagram (ERD) should reflect the implemented data structure and the relationships identified below.

The following key entities are maintained within the system:

- **Customer, Staff, and Admin Entities:** Customer accounts, staff accounts, and administrator accounts are maintained in separate collections. Authentication credentials are stored as password hashes, while role-based access control restricts access to protected system functionality.

- **Inventory Entity:** Contains branch-specific product information, including item name, category, brand, price, quantity, and image references. Inventory records are identified using database identifiers and associated with their respective branches.

- **Branch Entity:** Maintains information about physical business branches and supports branch-specific inventory and operational access.
- **Doctor and DoctorSession Entities:** Doctor records maintain information about available medical professionals. Doctor Sessions define examination schedules, assigned Doctors, branch locations, and patient capacities.

- **Appointment Entity:** Maintains customer eye examination bookings, including references to the relevant customer, Doctor Session, Doctor, and branch. Appointment records also contain booking status and queue-position information.

- **Prescription Entity:** Stores clinical optical prescription details linked to customer records. Prescription information is recorded and accessed according to authorized clinical permissions.

- **Cart, Order, and OrderItem Entities:** Maintain customer shopping carts, submitted orders, purchased items, order amounts, and order status histories. Order items preserve relevant purchase-time information for financial and inventory consistency.

- **Payment Entity:** Maintains payment transactions associated with customer orders, including payment status and verified payment references. Sensitive payment card credentials are not stored by the OSMS.

- **LoyaltyTransaction Entity:** Records eligible customer loyalty point earnings and applicable refund reversals. These records support the calculation and traceability of customer loyalty balances.

- **ReorderAlert Entity:** Maintains low-stock alerts associated with branch inventory records according to the configured organization-wide reorder threshold.

- **AuditEvent Entity:** Stores protected, append-only audit records for relevant inventory, financial, clinical, and administrative operations.

Additional supporting collections are used for system functionality, including chatbot usage tracking, knowledge records, notifications, and related operational information.

17.3 Data Integrity and Retention

- **Data Consistency:** Inventory quantities and related operational records are updated through authorized backend operations. The system uses validation rules, database references, indexes, and transactional controls to maintain data consistency.

Checkout-related stock reservations are processed using MongoDB transactions. Automated stock transfers between branches are not supported in the current implementation.

- **Security:** Sensitive information, including customer details and clinical prescriptions, is protected through authentication, role-based authorization, and secure HTTPS communication. Passwords are stored using Bcrypt hashing. Encryption-at-rest configuration and overall regulatory compliance remain subject to independent verification.

- **Backup and Recovery:** Database backup and recovery procedures are required to protect operational data against system failures and data loss. However, automated production backup schedules, restoration procedures, and successful recovery testing have not been independently verified and remain outstanding operational requirements.

- **Archiving and Retention:** The approved audit-trail architecture specifies a minimum seven-year retention period for critical audit records under ADR-012. The current implementation maintains append-only audit records without automatic TTL deletion. Full seven-year operational retention and archival guarantees remain subject to verification. Retention requirements for other business and clinical records must follow applicable approved policies and legal obligations.

- **Temporary Data Expiration:** Temporary chatbot usage records are designed to expire through MongoDB's Time To Live (TTL) mechanism. Correct operation of the TTL index in the production database must be independently verified.

18. Alternative Solutions Considered

Several strategic and technical approaches were considered during the planning of the Optical Shop Management System (OSMS) for Darshana Opticals (Pvt) Ltd. The alternatives were evaluated based on business requirements, development complexity, maintainability, and suitability for retail and clinical operations.

The web-based MERN stack solution was selected as the preferred approach.

18.1 Maintaining the Existing Manual System

The continuation of manual record-keeping, inventory tracking, and physical appointment scheduling was considered.

- **Reason for Rejection:** This approach was not selected because manual processes are more vulnerable to recording errors, delays in information retrieval, and difficulties in maintaining consistent inventory information across branches. It would also limit online shopping, digital appointment booking, and centralized business reporting.

18.2 Development of a Native Desktop Application

The development of standalone desktop software installed at individual branches was considered.

- **Reason for Rejection:** A desktop-based solution would require additional installation, maintenance, and update processes across branch locations. Providing centralized data access and online customer services would also require additional web-based components. A centralized web application was therefore considered more suitable for the project's requirements.

18.3 Utilization of Off-the-shelf Retail Software

The adoption of existing retail management or Point-of-Sale (POS) software was considered as an alternative to custom development.

- **Reason for Rejection:** General-purpose retail systems may require additional customization or integrations to support the project's specific requirements, including clinical prescription management, capacity-based Doctor Sessions, branch-specific inventory operations, loyalty point management, and AI chatbot assistance. Developing a customized system was preferred to provide greater control over these business workflows and future modifications.

18.4 Alternative Technology Stacks (e.g., PHP/MySQL)

Alternative web development technologies, including the LAMP stack (Linux, Apache, MySQL, and PHP), were considered.

- **Reason for Rejection:** Although these technologies are capable of supporting reliable web applications, the MERN stack was selected because it provides a JavaScript-based development environment across the frontend and backend. React.js supports the development of a component-based, interactive user interface, while Node.js, Express.js, and MongoDB support the project's selected application and data-management architecture. This approach was considered suitable for the team's development requirements, available skills, and implementation needs.

19. Feasibility Study

The feasibility of the Optical Shop Management System (OSMS) is assessed across four key dimensions: technical, economic, operational, and schedule feasibility. This

assessment considers the original project objectives, selected technologies, implemented functionality, and relevant project constraints.

19.1 Technical Feasibility

The technical feasibility of the project is supported by the following factors:

- **Technology Stack:** The selected MERN stack (MongoDB, Express.js, React, and Node.js) provides the necessary technologies for developing a centralized webbased application. The availability of documentation, development tools, and supporting libraries makes the approach suitable for the project.

- **Development Expertise:** The development team has demonstrated the ability to implement RESTful APIs, JWT-based authentication, role-based access control, database operations, and responsive user interfaces through the developed system.

- **Infrastructure:** The system uses Vercel for frontend hosting, Render.com for backend hosting, and MongoDB Atlas for database services. This cloud-based architecture supports remote access and centralized data management. Actual performance, availability, and scalability remain dependent on hosting resources and service limitations.

- **Integration:** The system incorporates external services, including PayHere for payment processing, Cloudinary for inventory image storage, and Gemini AI for chatbot assistance. SMTP-based email functionality is available for supported notifications. Automated SMS appointment reminders are excluded from the approved scope.

19.2 Economic Feasibility

The economic feasibility of the project is assessed based on expected development costs, operational benefits, and potential business value.

- **Development Costs:** The use of widely available development technologies and libraries reduces the need for expensive proprietary software licenses. However, cloud hosting and third-party services may introduce recurring operational costs depending on usage and service plans.

- **Long-term Savings:** The digitalization of inventory management, appointment scheduling, order processing, and reporting is expected to reduce manual work, improve data accuracy, and increase operational efficiency.

- **Revenue Growth:** The online storefront provides opportunities to reach a wider customer base and support additional product sales. Customer engagement features, including loyalty point management and chatbot assistance, may contribute to customer retention. Actual revenue growth has not been independently measured.
- **Return on Investment (ROI):** The project is expected to provide long-term operational and business benefits through improved efficiency and digital customer services. However, a financial ROI calculation has not been validated using actual operating costs, revenue changes, and measured savings.

19.3 Operational Feasibility

The operational feasibility of the system is assessed based on its compatibility with existing business processes and user requirements.

- **Workflow Alignment:** The system supports the digitalization of key business operations, including product sales, inventory management, eye examination appointments, prescription recording, and business reporting. These functionalities are designed to support the existing retail and clinical workflows.

- **User Acceptance:** The application provides role-based interfaces intended to support customers and staff with different levels of technical experience. Appropriate user training and usability evaluation remain important for successful adoption. Final user acceptance must be established through documented testing and stakeholder feedback.

- **Management Support:** The project is developed to address the operational requirements of Darshana Opticals (Pvt) Ltd. Continued management participation, feedback, and approval are important for successful system adoption and future maintenance.

19.4 Schedule Feasibility

The schedule feasibility of the project is assessed based on its original development plan, phased implementation approach, and available project resources.

- **Project Timeline:** The project was initially planned within a six-month development schedule, covering requirement analysis, system design, implementation, testing, and deployment.

- **Phased Development:** The project was divided into development sprints and functional modules, including user management, online shopping, inventory management, appointment scheduling, and administrative operations. This approach supported incremental implementation, testing, and review.

- **Resource Allocation:** Development responsibilities were distributed among team members to support the completion of assigned modules and integration activities. Final delivery readiness depends on completing outstanding system verification, documentation, and acceptance activities. Compliance with every original schedule milestone has not been independently established.

20. Risk Analysis

Potential risks affecting the successful operation, security, and maintenance of the Optical Shop Management System (OSMS) are identified and evaluated in this section.
The Risk Register outlines the likelihood, impact, risk level, and proposed mitigation strategies for each identified risk. The original risk classifications are preserved, while mitigation strategies are updated to reflect implemented controls, approved decisions, and outstanding verification activities.

Risk Register Table

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
RSK-001	Data may be lost or corrupted during the migration of existing manual records into MongoDB.	Technical	Low	High	Medium	Validate existing records before migration, perform incremental data imports, and maintain verified backups. Migration and recovery procedures must be tested before transferring critical business data.

RSK -002	Payment gateway integration issues or external service failures may delay checkout processing and release verification.	Schedule	Medium	High	High	Use the implemented PayHere integration with secure payment notification verification, controlled error handling, and transaction monitoring. Test payment scenarios using the approved sandbox environment and document unresolved integration failures.
RSK -003	Key development personnel may become unavailable during critical project activities.	Resource	Low	Medium	Medium	Maintain updated technical documentation, use shared version-control repositories, conduct knowledgesharing activities, and distribute essential technical knowledge among team members.

RSK -004	Functional requirements may change after the original requirements approval, affecting implementation scope and project schedules.	Requirements	Medium	High	High	Manage requirement changes through documented issues, architectural decision records, and team reviews. Evaluate and record the impact of approved changes on implementation , testing, and final project documentation.
RSK -005	Unauthorized users may attempt to access sensitive customer	Technical	Low	High	Mediu m	Apply JWTbased authentication, role-based access control, Bcrypt
	information or clinical prescription records.					password hashing, input validation, and HTTPS communication . Conduct security testing against relevant OWASP risks and address identified vulnerabilities before final acceptance.

RSK -006	Staff members may experience difficulties adopting the system because of limited technical experience or unfamiliar workflows.	Organisational	Medium	Medium	Medium	Provide useroriented interfaces, conduct appropriate staff training, prepare user guidance, and collect stakeholder feedback through usability testing and User Acceptance Testing (UAT).

Table 06: Risk Register of the Optical Shop Management System

Risk Assessment Note: The likelihood, impact, and risk levels above are retained from the original approved Risk Register. Mitigation activities that have not been independently verified must be considered planned or ongoing rather than completed.

21. References

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[8] Darshana Opticals OSMS Project Team, *OSMS Frontend Source Code*, GitHub Repository, 2026. Available: https://github.com/darshana-opticals-osms/frontend

[9] Darshana Opticals OSMS Project Team, *OSMS Infrastructure and Deployment Documentation*, GitHub Repository, 2026. Available: https://github.com/darshana-opticalsosms/infra

Client Requirement Sign-off Page

Client Representative Sign-off

Name: _____

Title: _____

Organisation: _____

Date: _____

Signature: _____

Project Supervisor Sign-off

Name: _____

Date: _____

Signature: _____

Note: Any changes to the approved requirements after this sign-off must be submitted as a formal Change Request and re-approved by both parties before implementation.